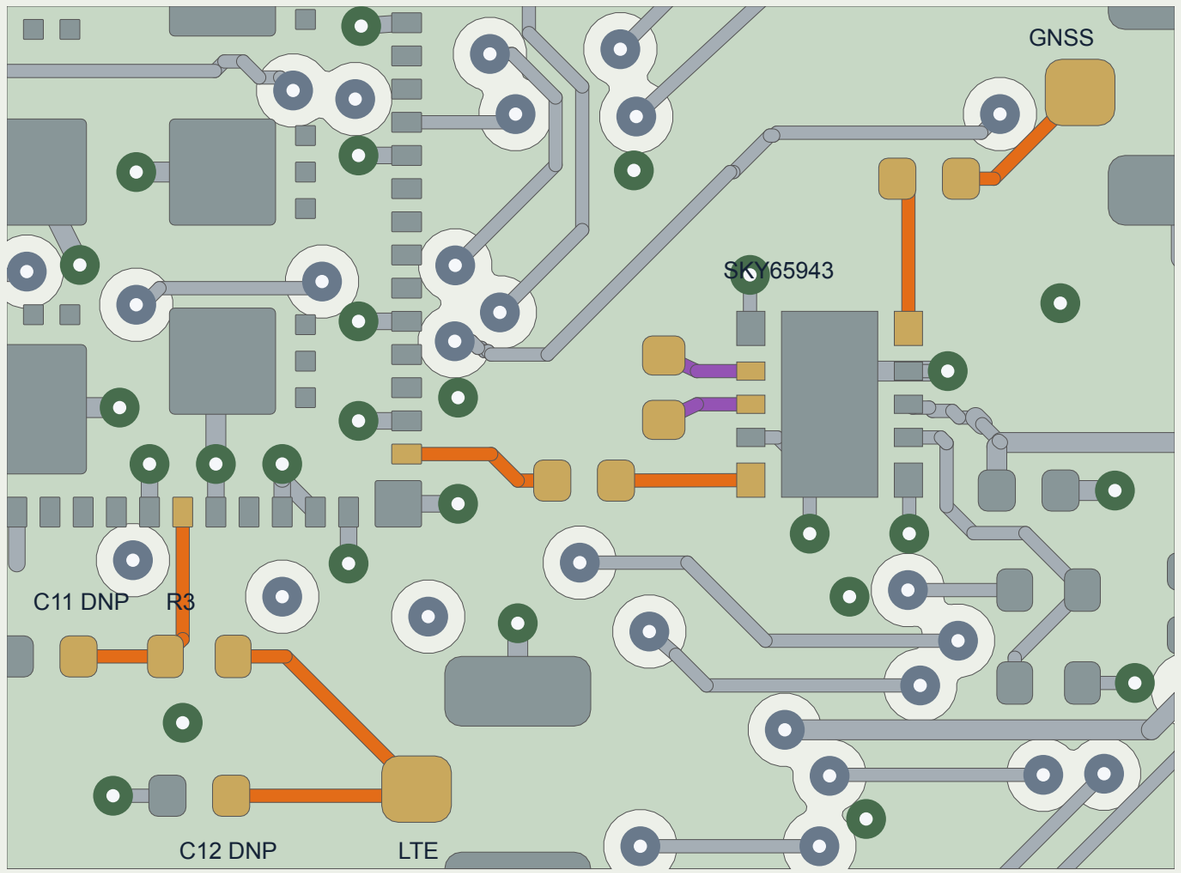
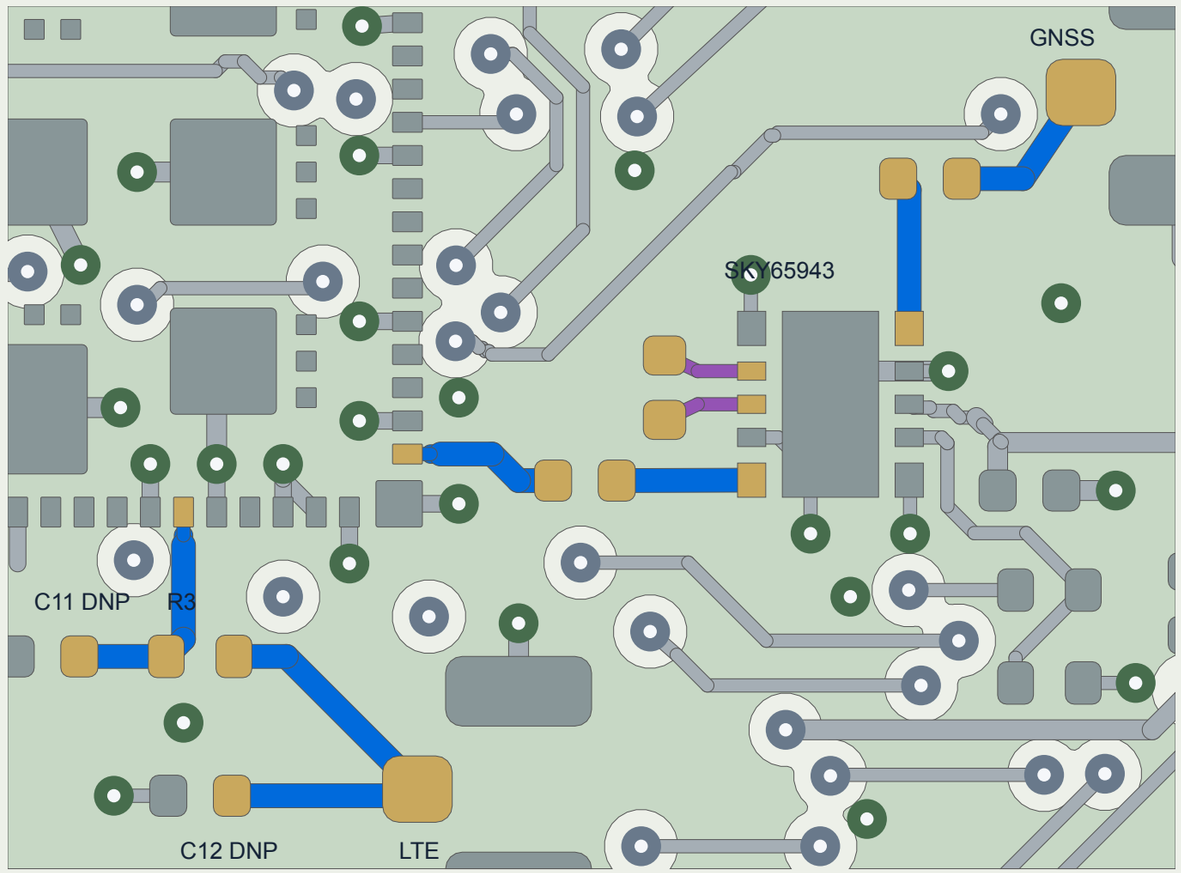
Collar tracker — RF geometry revision r05i

Native checks and two JLC PCB analyses pass. Electrical / RF / assembly / IP68: HOLD.

r05h — original 0.20 mm routes



r05i — 0.36 mm main routes



Changed: six port nets, two bounded module escapes, GNSS elbow, explicit candidate stack. Pads/vias/digital copper unchanged.

Reference: 1,761 native Contains probes agree; all revised RF trace copper projects onto In1.Cu GND.

Candidate: JLC06081H-7628; F.Cu → In1.Cu gap 0.2104 mm. Calculator: 0.3586 mm for nominal 50 Ω.

Limits: live calculator and guide disagree on Er / etch / mask; supplier confirmation and RF measurement remain open.

Module-pad GND retained: VBAT runs underneath on In2.Cu. Opening it without rerouting would expose the RF pads.

0.20/0.28 mm escapes, DNP matching stubs and FEM matching remain discontinuities requiring RF review.

Fit: 69 populated envelopes pass at 0.71 / 0.81 / 0.84088 / 0.91 mm. Original carrier interference is still detected.

No physical powered, antenna/body, factory-assembly or water/dust acceptance. Heart-rate hardware is not implemented.

PCB SHA256 0db809c4ecefbf577db1001164886709708f02ab6ba54f7a572e948583d726a1